MILITARY
SPECIFICATION

MIL-B-11453A(SigC)
3 September 1958
Superseding MIL-S-11453(SigC
14 September 1951

BATTERY, STORAGE BB-401( )/U

1. SCOPE

1.1 Scope.- This specification covers the requirements for one type of 24 cells rechargeable nickel-cadmium battery designated as Battery, Storage BB-401( )/U.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, and drawings of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

UU-T-111	Tape; paper, gummed (sealing and securing)
LLL-F-291	Fiberboard; corrugated, single face (flexible)
PPP-B-621	Boxes, wood, nailed and lock-corner
PPP-B-636	Boxes, fiber
PPP-T-76	Tape, pressure-sensitive adhesive, paper, water resistant

MILITARY

MIL-P-116	Preservation, methods of
MIL-B-117	Bags, interior, packaging
MIL-T-7928	Terminals; lug, crimp style, copper insulated
MIL-M-11473	Magnetic-particle inspection; soundness requirements for weldments
MIL-W-12683	Welding; joint design, data for (structures other than armor)
MIL-M-13231	Marking of electronic items
MIL-C-17435	Cushioning material, fibrous glass

STANDARDS

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MIL-STD-105	Sampling procedures and tables for inspection by attributes
MIL-STD-129	Marking for shipment and storage

DRAWINGS

SIGNAL CORPS

SC-C-33073	Form to be used for Pilot Pack Process Sheet
SC-DL-90967	Battery, Storage BB-401/U Drawing and Data List

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(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

2.2 Other publications.- The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN TRUCKING ASSOCIATION

National Motor Freight Classification No. A-3 and No. 14

(Application for copies should be addressed to the American Trucking Association, 1424 16th Street, N. W., Washington 6, D. C.)

ASSOCIATION OF AMERICAN RAILROADS

(1) Uniform Freight Classification No. 4

(Application for copies should be addressed to the Uniform Freight Classification, 202 Union Station, Chicago 6, Illinois)

(2) Consolidated Freight Classification Rules No. 21

(Application for copies should be addressed to Consolidated Committee, 1 Park Avenue and 33rd Street, New York 16, New York)

3. REQUIREMENTS

3.1 Construction.- The equipment shall be constructed in accordance with Drawing and Data List SC-DL-90967. In case of conflict between specification and drawings, the specification requirements shall govern.

3.2 Parts, materials, and processes; general.-

3.2.1 Selection.- Parts, materials, and processes shall be such as to enable the equipment to meet its performance and other requirements as specified herein.

3.2.2 Vents.- Each cell shall be equipped with a vent plug designed to prevent internal pressure from exceeding 2± 1.0 psi. The vent shall be air tight below 1 psi. (See 4.5).

3.3 Plastic material and parts.- Where not machined, plastic material and parts shall have the original smooth or polished surfaces. Surfaces that have been sawed, cut, punched, or otherwise machined shall be as smooth as practicable in accordance with good manufacturing practice for the intended application.

3.4 Crimping of solderless terminal lugs.- Crimping of solderless terminal lugs shall be so accomplished that the connections will meet the resistance (voltage drop) and tensile strength requirements as specified below and tests of Specification MIL-T-7928.

Test current - 32 amperes
Max. Voltage drop
  Initial - 6 millivolts
  After test - 11 millivolts
Min strength - 70 pounds

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3.5 Soldering.-

3.5.1 Flux and cleaning agents.- No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use. Flux for soldering of electrical connections shall be rosin, rosin and alcohol, or rosin and turpentine.

3.5.2 Process.- There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting action, and shall not be crystallized, overheated, or underheated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment; flux being spread over a larger area; or damage to the equipment. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.6 Cleaning.- Metal parts, after fabrication, shall be cleaned in accordance with good commercial practice, or as specified in an applicable document. Cleaning processes shall have no deleterious effect on the equipment. Corrosive material shall be removed completely before parts are assembled into the equipment. After assembly, components shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. If necessary, such cleaning shall also be performed before final assembly of components.

3.7 Welding.- Welds shall be of ample size. Where tack or spot welds are employed, the number of welds shall be sufficient to provide adequate strength and at least two welds shall be used to hold each welded part. Spot or tack welds in aluminum or magnesium shall be limited to lightly stressed, noncritical structures. Wherever practicable, the welded joint shall conform to Specification MIL-W-12683, and shall be such that grinding on the finished weld will be unnecessary.

3.7.1 Cleaning.- Surfaces to be welded shall be cleaned in accordance with good commercial practice and shall be free from rust, scale, paint, grease, and other foreign material. Welded assemblies shall be cleaned to remove scale, oxidation products, and excess flux by sandblasting, wire brushing, or other suitable means. Acid used for cleaning shall be completely neutralized and removed.

3.7.2 Process.- Preheating shall be employed where distortion is likely to result from welding. Welds shall have thorough penetration and good fusion and shall be free from scabs, blisters, abnormal pock marks, and other harmful defects. Where desirable internal stresses are likely to result from welding, the welded assembly shall be stress-relieved. Inert-gas shielded arc welding, shall be used, when practical for welding of aluminum or magnesium.

3.8 Electrolyte.- The electrolyte used in Battery, Storage, BB-401( )/U shall be a solution, approximately 31% by weight (1.300 to 1.310 specific gravity at 80°F) of chemically pure Potassium Hydroxide in distilled water.

3.9 Electrical requirements.- When discharged through a 2.8 ohms resistor in accordance with any of the discharge tests of section 4, the equipment shall maintain a potential ranging from 30.8 volts (maximum) to 25.2 volts (minimum) for not less than 15 minutes.

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3.10 Leakage.- There shall be no evidence of electrolyte leakage nor of electrical leakage when the battery is subjected to the leakage test of paragraph 4.13.

3.11 [illegible].- The equipment shall be capable of meeting the requirements of 3.9 after being charged with a trickle charge of 1.4 volts per cell for a period of one year at an average temperature range of from + 70°F to + 125°F as outlined in 4.11.

3.12 Weight.- The weight of the equipment shall be kept as light as practicable in keeping with the requirements of Drawing and Data List SC-DL-90967 and other requirements of this specification, and shall not exceed 13 pounds.

3.13 Service conditions.- The equipment shall meet the following service conditions:

3.13.1 Equipment nonoperating.- The equipment shall comply with the operational requirements of 3.13.2 after subjection to any of the following nonoperating conditions successively or in combinations encountered during worldwide short-term storage and transit:

(a) Temperature.- Exposure in the range of +160°F to -80°F.
(b) Elevation.- Elevation up to 25,000 feet above sea level.
(c) Orientation.- Indefinite storage in any orientation.
(d) Shock and acceleration.- As described in 3.14.

3.13.2 Equipment operating.- The equipment shall meet the performance requirements of this specification while subjected to any of the following conditions successively or in combinations likely to be encountered during worldwide operations:

(a) Temperature.- Ambient temperature in the range of +125°F to +70°F.
(b) Elevation.- Elevation up to 100,000 feet above sea level.
(c) Orientation.- Any orientation.

3.14 Shock and acceleration.- The equipment shall be capable of meeting the requirements of 3.9 after withstanding the shock and acceleration test of 4.8.

3.15 Marking.-

3.15.1 General.- Marking shall be in accordance with the requirements of Specification MIL-M-13231 except marking on plates specified in 3.15.2 shall be silk screening using material resistant to Potassium Hydroxide.

3.15.2 Plates.- Each battery shall be supplied with two plates, an identification plate, and an instruction plate. All marking shall be black on a stainless steel plate or equal.

3.15.2.1 Nameplate date of manufacture.- The date shown on the nameplate shall be that of the month and year of completion of the battery. The manufacturer shall be permitted to date, as of such month, batteries produced during the first three working days of the subsequent month, or the last three working days of the previous month.

3.15.2.2 Instruction plate.- Each battery shall bear, on the side opposite to that bearing the name plate, full instructions for operation and maintenance of Battery Storage BB-401( )/U.

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3.16 Preproduction samples.- The contractor shall furnish preproduction samples for approval, as required by the invitation for bids and contract. (see 6.1(c)(1).)

3.17 Technical literature, tools, and running spare parts.- Technical literature tools, and running spare parts shall be furnished as specified in the contract. Running spare parts shall be identical to corresponding parts in the items furnished on the order. (see 6.1.)

3.18 Workmanship.- The equipment shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs herein:

3.3 General requirements for plastic material and plastic parts.
3.5.1 Flux and cleaning agents for soldering.
3.5.2 Process for soldering.
3.6 Cleaning of equipment.
3.7.1 Cleaning for welding.
3.7.2 Process for welding.

4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of inspection.- Inspection shall be classified as follows:

(a) Preproduction inspection (does not include preparation for delivery). (see 4.2.)

(b) Procurement inspection. (Procurement inspection shall be the inspection performed by the contractor and by the Government, as specified by 4.3 and 4.17.)

(1) Procurement inspection of items before preparation for delivery. (see 4.3.)

(2) Procurement inspection of preparation for delivery. (see 4.17 and section 5.)

(c) Laboratory inspection. (Laboratory inspection is the inspection and tests performed at a Government Laboratory, as specified in 4.4)

4.2 Preproduction inspection.- This inspection will be performed by the Government unless otherwise specified in the contract. It shall consist of the preproduction inspection specified in table I and group A and group C inspection specified in tables II and III, respectively. Other nondestructive tests on preproduction samples may be performed to determine compliance with specified requirements.

Table I.- Preproduction inspection

Inspection	Rqt. Par.	Insp. Par.
Shock and acceleration	3.14	4.8

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4.3 Procurement inspection of items before preparation for delivery.- The minimum inspection performed by the contractor shall be as specified by (a) through (b) below--to demonstrate compliance with specified requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's records. In addition, the Government--at its discretion--will verify, by product inspection, the contractor's compliance with specified requirements. (See 6.5.)

(a) Procurement inspection shall consist of group A and group C inspection and shall be performed by the contractor as specified in 4.3.1 through 4.3.4 except that Group A inspection by the contractor shall be 100% inspection performed on each unit of production.

(b) When Standard MIL-STD-105 specifies actions by the Government, the Government may authorize the contractor to perform any of such actions except that responsibility for acceptance rests with the Government.

4.3.1 Group A inspection.- This inspection (including sampling) shall conform to table II and Standard MIL-STD-105 except that inspection by the contractor shall be 100% inspection (See 4.3(a)). Unless otherwise specified, normal inspection shall be used at the start of the contract. Group A inspection shall be performed in any order which is satisfactory to the Government.

Table II Group A Inspection

Inspection	Rqt. Par.	Insp. Par.	AQL Major	AQL Minor
Visual and Mechanical	3.18	4.14		
Cells (BB-403/U)			0.65%	2.5%
Complete Battery (BB-401( )/U			1.0%	4.0%
Electrical				*
Discharge Capacity	3.9	4.7	1.0%	
Leakage	3.10	4.13	1.0%	

*All electrical defects are considered major

4.3.2 Group B inspection.- Not required.

4.3.3 Group C inspection.- This inspection shall be as listed in table III.

4.3.3.1 Sampling for inspection of equipment.- Two batteries shall be selected without regard to their quality from each 250 units, or fraction thereof, that have passed group A inspection, for each group C inspection.

4.3.3.2 Sampling for inspection of solderless-terminal-lug connections.- Five specimens of each combination of wire and terminal shall be prepared each week for inspection. A specimen is a 1-foot length of wire on which a terminal lug has been applied, using the same tools and procedures used in crimping the terminal lugs in production of the items on contract.

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4.3.3.3 Noncompliance.- If a sample unit fails group C inspection, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken on the process and all units of product which were manufactured with the same conditions, materials, processes, etc. If the Government inspector does not consider that the corrective action will enable the product to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer. (see 6.4.)

Table III Group C Inspection

Inspection	Rqt. Par.	Insp. Par.
Altitude Test	3.13.2	4.9
Discharge at High Temperature	3.13.2	4.10
Crimping of Solderless Terminal Lugs	3.4	4.15
Welding	3.7	4.16

4.3.4 Disposition of nonconforming product.- The following shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means employed to correct the fault:

Sample units found defective during group A, B, or C inspection.

Inspection lots which failed during group A or B inspection.

Product which has been reworked as a result of failure during group C inspection.

The required information shall be presented to the Government when the product is submitted and shall become the property of the Government.

4.4. Government laboratory inspection.- This inspection shall be as listed in table IV. (See 6.6.)

Table IV Laboratory Inspection

Inspection	Rqt. Par.	Insp. Par.
Storage Test	3.13.1	4.11
Stand-by Life Test	3.11	4.12

4.4.1 Number of samples.- The total number of samples to be selected from product and furnished to the Government laboratory shall be as follows:

Total number of batteries on release	Total number of samples to be selected
20-100	2
101-500	4

Plus 4 additional batteries for each additional 500 batteries, or part thereof, on order

4.5 Vent plug test.- The vent plug shall be tested to determine compliance with requirements specified in 3.2.2 by installing it in an adapter to which a variable pressure of compressed air is connected. The adapter shall then be immersed in water and the air pressure adjusted to $1^{+0}_{-.1}$ psi. If emission of air bubbles [illegible] yrs, the vent plug has failed the test. The air pressure shall then be adjusted to $3^{+.1}_{0}$ psi. [illegible] bubbles does not occur, the vent plug has failed the test

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4.6 Test conditions.- After being fully charged in accordance with the directions printed on the instruction label, the batteries shall be stored at 75 ±7°F and at 50 ±5% R.H. for 16 to 48 hours. The batteries shall then be subjected to the tests specified in 4.7, 4.8, and 4.9.

4.7 Full Charge capacity discharge.- After being conditioned as specified in 4.6, the equipment shall be discharged, at 75 ±7°F and at 50 ±5% R.H., through 2.8 ohms to an end voltage of 25.2 volts. After each discharge, unless otherwise specified, each Battery, Storage, BB-401( )/U shall be recharged in accordance with the directions printed on the instruction plate. (3.15.2.2.)

4.8 Shock and acceleration test.- After being conditioned as specified in 4.6, the equipment shall be subjected to applications of shock and acceleration of the magnitude and direction, and in the order listed below:

a. Along the plane at right angle to the 4.018" x 6.605" faces of the battery.

(1) 50 G acceleration for 3 seconds.

(2) 3 G accelaration for 26 seconds.

(3) 100 G for 6.5 milliseconds as imparted by the Medium Impact Machine, Type 150 VD, manufactured by the Barry Corporation, Cambridge, Massachusetts, or equal.

b. Along the plane at right angle to the 7.38" faces of the battery.

(1) 10 G accelaration for 150 seconds.

(2) 100 G for 7.5 milliseconds as imparted by the Medium Impact Machine, Type 150 VD, manufactured by the Barry Corporation, Cambridge, Massachusetts, or equal.

c. A centrifugal acceleration to produce rotation at 600 rpm for 10 seconds about the axis perpendicular to the 4.018" x 6.605" faces.

These tests shall be conducted at 75 ± 7°F., 50 ± 5% R.H. and shall be followed by a discharge through 2.8 ohms at an end voltage of 25.2 volts.

4.9 Altitude test.- After being conditioned as specified in 4.6, the equipment shall be placed in a pressure chamber and the pressure reduced to 8 ±0.8 mm of mercury. The temperature of the chamber at the beginning of the test shall be 75± 7°F; during the remainder of the test, the temperature of the chamber shall be uncontrolled. The pressure, 8mm of mercury, shall be reached in not less than 10, nor more than 15 minutes and shall be maintained for 25 minutes, during which time the equipment shall be discharged through 2.8 ohms to 25.2 volts. Pressure shall then be raised to ambient atmospheric pressure within 10 minutes. Half of the number of batteries subjected to this test shall be discharged while in an inverted position. Following this test, the electrolyte level of the equipment shall be adjusted and the equipment shall be resubjected to the test specified in 4.7.

4.10 Discharge at high temperature.- After being fully charged, the batteries shall be allowed to stand at 125 ± 2°F. for 12 hours, then discharged at 125 ± 2°F. through 2.8 ohms to an end voltage of 25.2 volts. Half of the number of batteries subjected to this test shall be discharged while in the inverted position. The test shall be followed by a repetition of the test of 4.7.

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4.11 Storage test at Government Laboratory. The batteries shall be discharged in accordance with 4.7 except that the discharge shall be continued to 0 volts. The batteries, in the discharged condition, shall be stored with normal maintenance as specified by the instruction plate (3.15.2.2) for two years in accordance with the following schedule.

3 days at -80°F.
120 days at 0°F.
480 days at 80°F.
120 days at 111°F.
3 days at 160°F
4 days at 80°F.

After this storage period the batteries shall be tested in accordance with 4.7.

4.12 Stand-by life test at Government Laboratory. The batteries, after being charged to a full charge, shall be maintained on a trickle charge at 33.6 ±0.3 volts for one year at a temperature of 75± 7°F, 50 ±5% R.H., with normal maintenance as specified by the instruction plate (3.15.2.2). At the end of the one year period the batteries shall be tested in accordance with 4.7.

4.13 Leakage.-

4.13.1 Electrical leakage test for cells.- Each cell of the equipment shall be tested for electrical leakage by immersing the cell in water up to and slightly over the sealing line between the container and cover. Care must be exercised that no water comes in contact with the top of the cover or the terminals. The cell shall be kept immersed in this manner for a minimum of fifteen (15) seconds, after which a reading shall be taken with an ohm meter (R x 10,000 ohm) between the water and one (1) of the cell terminals. Any reading on the ohm meter shall be cause for rejection of the cell.

4.13.2 Electrical leakage test for battery.- The entire battery, in a charged condition, shall be tested for electrical leakage by measuring the voltage between each terminal and the metal case of the battery.

4.13.3 Electrolyte leakage test for cells.- The outside surfaces of each cell shall be wiped dry by means of a dry cloth or dried by means of a blast of warm air. The cell shall then be rolled over on all four sides on dry blotter paper. Any wet spots appearing on the blotter paper shall be cause for rejection of the cell.

4.14 Visual and mechanical inspection.- The equipment shall be given a visual and mechanical inspection to determine that the dimensions, markings, and workmanship are in accordance with the requirements of this specification.

4.15 Testing of solderless terminal-lug connections.- Connections made with solderless terminal-lugs shall be tested to meet the requirements of 3.4.

4.16 Welding inspection.- Welding shall be inspected by the magnetic particle method specified in Specification MIL-M-11473 and by the ultrasonic method using a Reflectoscope, as made by Sperry Products, Incorporated, Danbury, Connecticut, or equal. Both tests shall be performed on each sample unit.

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4.17 Procurement inspection of preparation for delivery.- The minimum inspection performed by the contractor shall be as specified in 4.17.1 and 4.17.2. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's records. In addition, the Government - at its discretion - will verify, by product inspection, the contractor's compliance with specified requirements. (See 6.5.) Such verification shall not exceed the amount of inspection required herein of the contractor. When Standard MIL-STD-105 specifies actions by the Government, the Government may authorize the contractor to perform any of such actions except that responsibility for acceptance rests with the Government.

4.17.1 Preservation and packaging.- Inspection of preservation and packaging shall be as specified in Specification MIL-P-116. Classification of defects shall be as shown on table V.

TABLE V

PRESERVATION, AND PACKAGING AND MARKING THEREOF

MAJOR	MINOR
1. Use of improper or defective material.	1. Item not properly blocked or braced within the unit package to prevent movement.
2. Quantity in unit package not as specified.	2. Packaging material damaged.
3. Incorrect packaging method applied.	3. Any item of marking information other than 7 thru 9 omitted, incorrect, or illegible.
4. Cushioning or padding omitted.	
5. Cushioning inadequate for the physical and mechanical protection of the item.	
6. Unsealed, punctured, or improperly sealed barrier, bag, wrap, or envelope.	
7. Stock number omitted, incorrect, or illegible.	
8. Nomenclature omitted, incorrect, or illegible.	
9. Marking of quantity of items in package omitted, incorrect or illegible.	
10. Different stock numbered items in the same unit package.	

4.17.2 Packing and marking of exterior containers.- Packing and marking of exterior containers shall be given visual inspection for the defects listed below and to determine conformance with the approved process sheet furnished by the contractor as required by the contract. This inspection shall conform to the Appendix to Standard MIL-STD-105. Inspection level L-8 shall be used for normal inspection and L-6 for reduced inspection. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. The reduced inspection procedure shall be R-1. The AQL for major defects shall be four percent and the AQL for minor defects shall be ten percent. Classification of defects shall be as shown on table VI.

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TABLE VI

PACKING AND MARKING THEREOF

MAJOR	MINOR
1. Use of improper or defective material.	1. Unsealed carton.
2. Quantity in pack not as specified.	2. Defective taping or sealing of carton.
3. Box closure not as specified.	3. Any other box defect which may be considered minor by definition of Standard MIL-STD-105.
4. Type, grade, class, and style of the shipping container not as specified.	4. Any item of required marking information other than 8 thr 14 listed under major defect omitted, incorrect, or illegible.
5. Strapping omitted.	
6. Strapping inadequate or incorrectly applied.	
7. Shipping documents or packing list omitted.	
8. Stock number omitted, incorrect or illegible.	
9. Nomenclature omitted, incorrect or illegible.	
10. Marking of quantity of items in pack omitted, incorrect, or illegible.	
11. Destination marking omitted, incorrect, or illegible.	
12. Service designation (color marking) omitted.	
13. Specified special marking and labeling such as NAP labels, shipment digit markings, etc., omitted, incorrect, or illegible.	
14. Overseas code marking omitted, incorrect, or illegible.	

4.17.2.1 Inspection lot.- A lot for visual inspection of the pack shall be all completed packs which are identical and will be submitted for acceptance at one time.

4.17.2.2 Procedure in case of failure.- If an inspection lot fails, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken. If the contractor and Government inspector cannot agree on the effectiveness of the corrective action, the matter shall be referred to the contracting officer for resolution.

4.17.2.3 Disposition of nonconforming product.- Disposition of nonconforming product (sample units and inspection lots) shall be in accordance with the requirements of Standard MIL-STD-105 for disposition of rejected product. When submitted for acceptance, such product shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means employed to correct the fault. The required information shall be presented to the Government when the product is submitted and shall become the property of the Government.

4.18 Rough handling test (preparation for delivery).- When rough handling test is required by the contract (see 6.1 (c)), the following functional tests shall be conducted to determine freedom from operational malfunction caused by the rough-handling.

4.7 Full Charge Capacity Discharge

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4. [illegible]

5. PACKAGING

5.1 Preservation and packaging.-

5.1.1 Level A.- Battery, Storage BB-401( )/U shall be preserved, packaged and [illegible] as specified for the designated methods as prescribed in Specification MIL-P-116 and described in the following paragraphs.

5.1.1.1 Technical manual.- Each technical manual shall be packaged Method IC-3 as follows: Wrap the each technical manual within a close-fitting type I, class b bag, conforming to Specification MIL-B-117. Heat seal the bag to effect closure.

5.1.1.2 Battery Storage BB-401( )/U.-

5.1.1.2.1 Prepackaging.- The solid plugs (shipping screws) with gasket, shall be positioned over each cell and turned down firmly to preclude leakage.

5.1.1.2.2 Vent plugs and rubber gasket.- Twenty-five each, vent plugs and rubber gaskets shall be packaged Method IC-3 as follows: Group the vent plugs and the rubber gaskets separately. Cushion each group by wrapping in type I, single faced, flexible, corrugated fiberboard, conforming to Specification LLL-F-291. Secure the cushioning with class 1, gummed paper tape, conforming to Specification UU-T-111. Place the cushioned vent plugs and rubber gaskets together within a close-fitting, type I, class b bag, conforming to Specification MIL-B-117. Heat seal the bag to effect closure. Secure the bagged items to one of the lifting eyes of the battery with pressure-sensitive, waterproof tape, conforming to Specification PPP-T-76.

5.1.1.2.3 Battery, Storage BB-401( )/U:- Four each, battery shall be packaged Method IC-5 as follows: Cushion each battery individually on all surfaces with pads fabricated of fibrous glass cushioning material, type II, having a minimum thickness of one inch, conforming to Specification MIL-C-17435. Place four each cushioned batteries within a close-fitting RS style, grade 3 (V3c), CF box, conforming to Specification PPP-B-636. Box closure shall be in conformance with the box specification. In addition seal all seams and joints with pressure-sensitive waterproof tape, conforming to Specification PPP-T-76.

5.1.2 Level C.- Battery, Storage BB-401( )/U shall be packed in accordance with commercial practice, and in a manner that will afford protection against corrosion, deterioration and physical damage during direct shipment to the first receiving activity.

5.2 Packing.-

5.2.1 Level A.- Battery, Storage BB-401( )/U shall be packed for shipment as follows: Place four each battery packaged as specified in 5.1.1.2.3 within a close-fitting, style 4, class 2, nailed wood box, conforming to Specification PPP-B-621. Place technical manual (see 5.1.1.1) between the contents and lid of the box. Box closure shall be in conformance with the box specification.

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5.2.1.1 Metal strapping.- Shipping containers shall be strapped in conformance with the appendix of the applicable container specification only for direct shipment to ports.

5.2.2 Level B.- Battery, Storage BB-401( )/U shall be packed for shipment as specified in 5.2.1 except the style 4 box shall be class 1.

5.2.3 Level C.- Battery Storage BB-401( )/U shall be packed for shipment, conforming to the requirements of Uniform Freight Classification No. 4 and Consolidated Freight Classification Rules 21 for rail shipment, National Motor Freight Classification No. A-3 and No. 14 for truck shipment, and the regulations of other carriers as applicable to the mode of transportation employed at lowest transportation rates.

5.3 Marking.- Interior packages and exterior shipping containers shall be marked in accordance with the applicable provisions of Standard MIL-STD-129.

5.3.1 Special marking.- In addition to the requirements specified in 5.3, shipping containers shall be marked in conformance with the applicable requirements of Interstate Commerce Commission Regulations for Transportation of Explosives and other dangerous articles.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendment thereto.
(b) Level of packaging and level of packing required for shipment. (Level A, level B, and level C.)
(c) Preproduction inspection:

(1) Three samples of each item cited in section 1 are generally required so that lengthy environmental tests can be completed on one sample while complete performance measurements can be made on the other samples. (See 3.16)
(2) Preproduction pack(s) as follows:

a. Makeup of pack(s).
b. Number of each kind of pack to be submitted.
c. Inspection to be performed thereon--including rough handling test, which will not be performed as procurement inspection.
d. Five copies of process sheet, using format shown on Drawing SC-C-33073 and accompanied by such drawings, sketches, and other data as may be necessary to completely describe the procedure followed in fabrication of the preproduction pack(s) and to identify the items therein. These data should be submitted to the contracting officer with the preproduction pack.
e. A packaging technician from the procuring agency will evaluate the preproduction pack(s).

(d) Marking and shipping of samples.
(e) Place of final inspection.
(f) Technical literature required. (See 3.17)
(g) Quantity of tools and running spare parts required. (See 3.17)

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6.2 Nomenclature.- The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: BB-401W/U. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

6.3 Location of operational inspection.- It is desirable that the operational inspection (4.7) be performed at a location that will minimize handling (which might cause damage to the equipment) after this inspection is completed. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

6.4 Group C inspection.- Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.3.3.3.)

6.5 Verification inspection.- The amount of verification inspection (by the Government) will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product, and will normally be identified by the categories listed below:

(a) Type A--The total of that inspection set forth in the Quality Assurance Provisions of this specification or the contract. Included in this category is that amount of inspection referred to as normal and tightened inspection by Military Standard 105.

(b) Type B--That inspection set forth in the Quality Assurance Provisions of this specification or the contract reduced in amount under the reduced inspection provisions of Military Standard 105.

(c) Type C--A reduced inspection procedure resulting in a material reduction in the amount of inspection set forth in the Quality Assurance Provisions of this specification. The amount of inspection is less than that provided for in type B and is based upon a consistently acceptable product resulting from a planned quality control system voluntarily employed by the contractor in the production of the product.

6.6 Government laboratory inspection.- The Government laboratory inspection, specified in Table IV, shall be conducted at a Government laboratory, for adjustment purposes, in accordance with a standard contract clause for storage battery requirements.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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